

PLUTO AND ITS MOONS

26–27 Celestial objects

38–39 Gravity, motion, and orbits

102–103 The family of the Sun

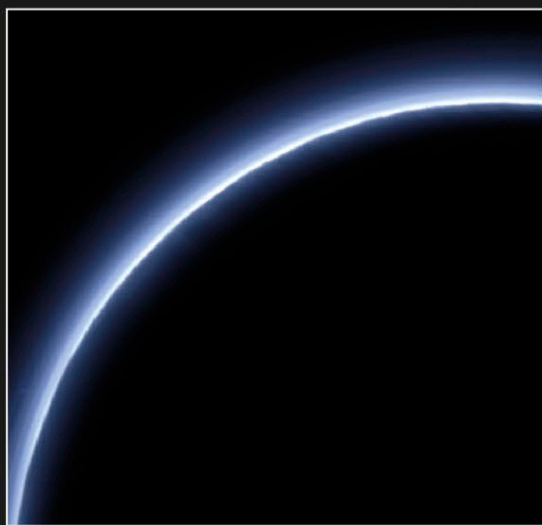
The Kuiper Belt and Oort Cloud 210–213

DISCOVERED BY US astronomer Clyde Tombaugh in 1930 (see p.209), Pluto was once classified as a planet in its own right. This outlier of the solar system was reclassified as a dwarf planet in 2006 and

is acknowledged as the first Kuiper Belt Object (KBO) to be discovered. It orbits closer to the Sun than other large KBOs but is probably not exceptional other than having been seen at close quarters by the New Horizons mission (see p.211) in 2015.

ORBIT

Pluto's orbit is inclined and eccentric. Its 248-year orbit ranges between about 2.7 billion miles (4.4 billion km) and 4.6 billion miles (7.4 billion km) from the Sun, meaning that Pluto sometimes lies closer in than Neptune (most recently between 1979 and 1999). However, the pronounced tilt of Pluto's orbit (at an angle of 17.1° to the ecliptic), combined with the fact that it completes precisely two orbits in the time it takes Neptune to complete exactly three orbits, makes close encounters between the two impossible.



BLUE SKY

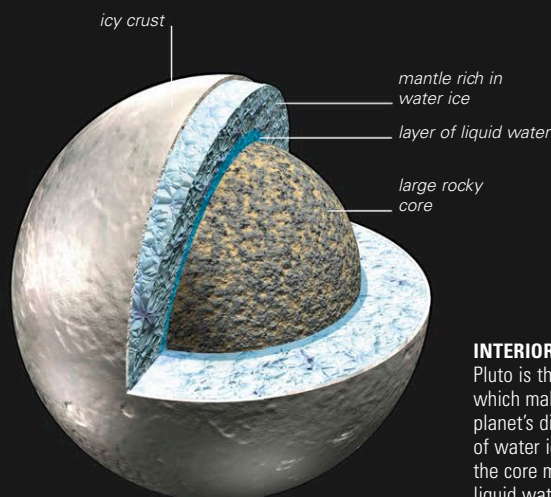
Looking back toward the Sun 3.5 hours after it had passed Pluto, from a range of about 120,000 miles (200,000 km) beyond, the New Horizons probe could see sunlight scattered and made blue by the photochemical haze in Pluto's atmosphere.

ATMOSPHERE

The existence of Pluto's atmosphere was confirmed in 1988, when the planet occulted a star and the starlight was cut off much more slowly than would have happened with an airless body. It is mostly nitrogen mixed with less than 1 percent of methane and a smaller proportion of carbon monoxide. The surface pressure of the atmosphere was found to be about 100,000 times less than the Earth's atmospheric pressure by New Horizons in 2015, and will probably decrease as Pluto's orbit carries it farther from the Sun, allowing more of its atmosphere to freeze onto the ground. The surface temperature is -351°F to -387°F (-213°C to -233°C), but the atmospheric temperature increases with height to a maximum of about -261°F (-163°C) at a height of about 19 miles (30 km) because of the absorption of sunlight by methane. This process leads to methane molecules linking together into hydrocarbon chains constituting smog particles, which slowly settle to the ground and make Pluto's atmosphere hazy.

STRUCTURE

Pluto is similar to a slightly smaller and less massive version of Neptune's largest moon, Triton (see p.207). Pluto's crust is mostly water ice, which is very strong and rigid under Pluto's low-temperature surface conditions. Pluto's density indicates that it must have a substantial rocky core. Radioactive heat generation within the core may be sufficient to melt the overlying ice to maintain an internal ocean of liquid water. The surface is covered mostly by nitrogen ice, with traces of frozen methane and carbon monoxide, too. Dark red discoloration of parts of the surface is probably caused by hydrocarbon haze particles that have settled to the ground.



INTERIOR OF PLUTO

Pluto is thought to consist of a rocky core, which makes up about 70 percent of the planet's diameter, surrounded by a mantle of water ice and a thin, icy crust. Heat from the core may help sustain a thin layer of liquid water between the core and mantle.